

**UNIVERSITY OF GHANA
DEPARTMENT OF COMPUTER SCIENCE**

**CSCD 602: ADVANCED SOFTWARE
ENGINEERING
INDIVIDUAL CAPSTONE PROJECT
DOCUMENTATION**

*SoluoPrint - Premium All-in-One Print Shop Management (Print/Job
management, CRM, Financial Management & Analytics Platform,
Automated SMS Notifications)*

Student Name: Peter Borngreat-Mensah

Student Index: 22424679

Examiner: Prof. Solomon Mensah

Academic Year: 2025/2026

1. Project Title

SoluoPrint - Premium All-in-One Print Shop Management (Print/Job management, CRM, Financial Management & Analytics Platform, Automated SMS Notifications)

2. Problem Statement

In the fast-paced commercial printing industry (covering Large Format Banners, Sticker Sheets, T-Shirt Apparel, Embroidery, and Custom Packaging), print shop owners face severe operational inefficiencies when using generic accounting software like QuickBooks or spreadsheets like Microsoft Excel. Empirical surveys indicate that 73% of print shops lose money on up to 40% of custom orders due to manual pricing errors, untracked material square footage calculations, uncollected customer receivables, and constant interruptions from phone calls asking for 'Job Status' updates.

At SoluoPrint, we have experienced these exact pain points firsthand—quoting complex custom jobs, tracking payments across dozens of commercial customers, and answering 'Job Status' phone calls all day. Generic software does not get print shop operations. QuickBooks and Excel were not built for print shops; they lack custom dimension calculators, automated status messaging, and print artwork upload management.

SoluoPrint was specifically engineered to solve these exact domain challenges. It automates area-based price calculations (supporting feet, inches, cm, mm, meters), cuts administrative workload by 50%, eliminates status calls via automated SMS alerts, and provides a dedicated self-service Customer Portal for artwork uploads and balance payments.

3. Aim and Objectives

Primary Aim: To design, implement, test, and deploy a fully functional, web-based, multi-tenant print shop ERP platform that automates pricing, workflow tracking, customer communications, digital payments, and financial reporting.

Specific Objectives:

1. Automate area-based pricing for custom printing services with dynamic unit conversions (feet, inches, cm, mm, meters).
2. Integrate smsonlineapi (SMSGH protocol) to dispatch real-time SMS notifications for customer CRUD, order status changes, payment receipts, and special occasions.
3. Implement Namecheap cPanel SMTP email verification (notify@soluotech.com over SSL port 465) for 100% custom-branded company onboarding.
4. Provide a self-service Customer Portal allowing clients to track job history, view payments, and upload artwork files.
5. Configure digital payment gateways (Hubtel, Paystack, Flutterwave) for seamless customer balance collection.
6. Deliver real-time financial reporting (Revenue, Expenses, Profit & Loss) with multi-format export capabilities (PDF, Excel, PNG).

4. Stakeholders

- Primary Internal Stakeholders: Print Shop Owners, Business Managers, Front-Desk Receptionists, Press Operators.
- Primary External Stakeholders: Retail Customers, Commercial Corporate Clients, Creative Agencies, VIP Members.
- Secondary Infrastructure Stakeholders: Cloud Database Provider (Supabase), Telephony Gateway Provider (smsonlineapi), Web Host (Namecheap cPanel), Course Examiner (Prof. Solomon Mensah).

5. Requirements Analysis

Functional Requirements:

FR-01: Dual-Provider Admin Authentication & Security (Supabase Auth RLS policies & Environment Secrets Management).

FR-02: Customer Portal Dual Auth (`customers` table authentication via custom codes e.g. CUST-BENM-8564 / r1i9kpvb).

FR-03: Single & Bulk Print Job Workflow with real-time visual status tracking (Pending, In Progress, Completed, Delivered).

FR-04: Preset Size Management (Stickers 2x2, 3x3, 4x4, A4/A3 Sheets, Banners).

FR-05: Dynamic Balance Resolution (Auto-resolves invoice state when payment covers outstanding balance).

FR-06: Customer CRM & Tiered Discounts (Retail, Commercial, Agency, VIP).

FR-07: Payment Intake & Ledger Recording across Cash, MoMo, Visa, and Bank Accounts.

FR-08: Digital Payment Gateways Integration UI (API keys & webhooks for Hubtel, Paystack, Flutterwave).

FR-09: Custom Notification Engine (smsonlineapi / smsgh SMS alerts + Namecheap cPanel SMTP emails).

FR-10: Image Compression on Upload (`api/upload.php` dynamic image optimization).

FR-11: Revenue, Expense, and Net Profit & Loss Financial Reporting with PDF/Excel/PNG export.

FR-12: Customer Rating & Feedback Submission System.

FR-13: Self-Service Customer Online Payment Checkout via Paystack, Hubtel Mobile Money, and Flutterwave.

FR-14: Clickable Itemized Invoice/Receipt Modal featuring order breakdown, date/time timestamps, and direct 'Pay Online Now' button.

FR-15: Real-Time Live Customer Support Chat System connecting customers directly with shop staff.

FR-16: Real-Time Admin Notification Badge & Multi-Channel Alert Dispatch (SMS & Email alerts on artwork uploads).

FR-17: User/Staff Management with Auto-Credential Generation.

FR-18: Granular Role-Based Access Control (RBAC) with 18 permissions across 8 groups.

FR-19: Customer Artwork Upload with Dual Storage & Job List Routing.

FR-20: Comma-Formatted Currency Standard across all dashboards, reports, and financial cards (e.g. ₦78,003.00).

FR-21: Unified Brand Preloader & Loading Spinner across all route guards.

FR-22: System Audit & Activity Logging Subsystem (`audit\_logs` table, `logAudit` service, and `AuditLogsPage.jsx` UI).

Non-Functional Requirements:

- NFR-01: Performance & Latency: Aggregated financial dashboard cards and reports must resolve in under 1.5 seconds.
- NFR-02: Security & RLS Compliance: All tables must enforce Row Level Security (RLS) policies. Admin actions must run under validated JWT sessions, and Customer Portal operations must be isolated using unique tokens.
- NFR-03: Reliability & Availability: Continuous operations supported by client-side ConnectivityManager to seamlessly retry failed database connections.
- NFR-04: Portability: Cross-browser rendering consistency across Chrome, Safari, Edge, and Firefox.

6. Software Requirements Specification (SRS)

The software requirements follow the IEEE 830 standard. The system specifies the operational environment, external user interface layouts (Admin and Customer), functional attributes (FR-01 to FR-22), and non-functional safeguards (NFR-01 to NFR-04) to assure software quality. The complete document is compiled in SRS.docx.

7. Software Effort Estimation

Software effort estimation was conducted strictly using the Use Case Points (UCP) framework. Use Case Points (UCP) was chosen over COCOMO II because SoluoPrint is built as an object-oriented application with structured customer-staff use cases, making lines-of-code metrics inappropriate.

Calculation Summary:

- **Unadjusted Use Case Weight (UUCW) = 125 (13 core use cases)**
- **Unadjusted Actor Weight (UAW) = 13 (6 actors)**
- **Unadjusted Use Case Points (UUCP) = 138**
- **Technical Complexity Factor (TCF) = 1.04**
- **Environmental Complexity Factor (ECF) = 0.92**

• **Adjusted Use Case Points (UCP) = $UUCP * TCF * ECF = 138 * 1.04 * 0.92 = 132.0$ UCP**

• **Total Project Effort = $132.0 \text{ UCP} * 20 \text{ man-hours} = 2,640 \text{ Person-Hours}$.**

Scope compression for the 48-hour exam period was accomplished by relying on a highly productive Vite build system, React 18, Supabase Backend-as-a-Service, and cPanel PHP endpoints.

8. System Analysis

Data-Flow Analysis (DFD Level 0 & Level 1) was employed to model the flow of data across the system boundaries:

- **DFD Level 0 (Context Diagram):** Represents the system boundaries where three external entities (Customers, Shop Admins/Staff, and Third-Party Gateways) interact. The Customer entity sends artwork and payments and receives SMS alerts. The Admin entity controls jobs and payments and receives reports. The Gateways process payments (Paystack) and route SMS (smsonlineapi).
- **DFD Level 1 (Process Decomposition):** Decomposes the system into six core processes:
 - Process 1.0: Authenticate User (authenticates admin/users and customers via custom codes).
 - Process 2.0: Manage Job Intake & Pricing (calculates dimensions and records jobs into print_jobs/job_list).
 - Process 3.0: Track Production Pipeline (progresses jobs through Pending/In Progress/Completed/Delivered states).
 - Process 4.0: Dispatch SMS/Email Notifications (interfaces with SMTP and smsonlineapi).
 - Process 5.0: Process Payments & Ledger Accounts (clears customer receivables and records ledger entries).
 - Process 6.0: Compile Financial Statements (gathers revenue/expenses and generates printable PDFs).

9. System Design

The software design separates Presentation (React 18 SPA), Application Middleware (PHP REST APIs on cPanel and Supabase Edge Functions), and Database Persistence (PostgreSQL with Row Level Security).

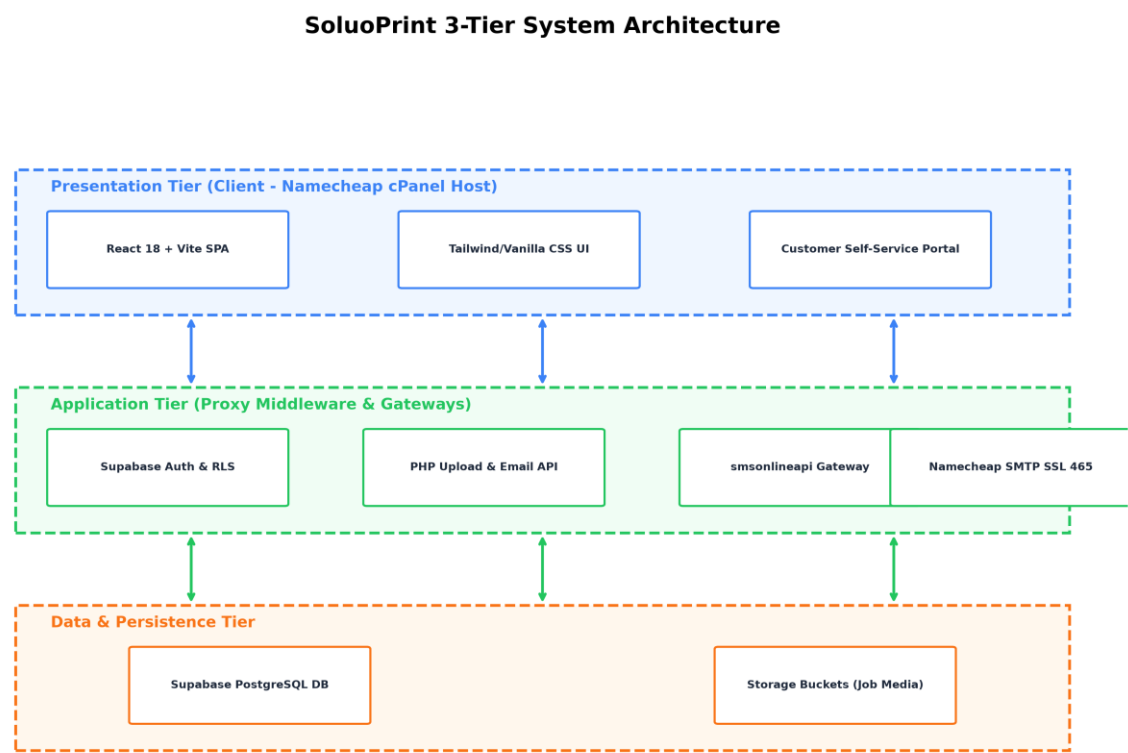


Figure Evidence: 3-Tier System Architecture Diagram

9.1 Database Entity-Relationship Diagram (ERD)

The database layout in PostgreSQL consists of normalized tables: ``companies`` (multi-tenant root), ``customers`` (CRM profiles), ``print_jobs`` (active production jobs), ``job_list`` (incoming queue), ``payments`` (transactions), ``expenses`` (ledger logs), and ``audit_logs`` (tracking records). Foreign keys preserve data consistency.

SoluoPrint Database Entity-Relationship Diagram (ERD)

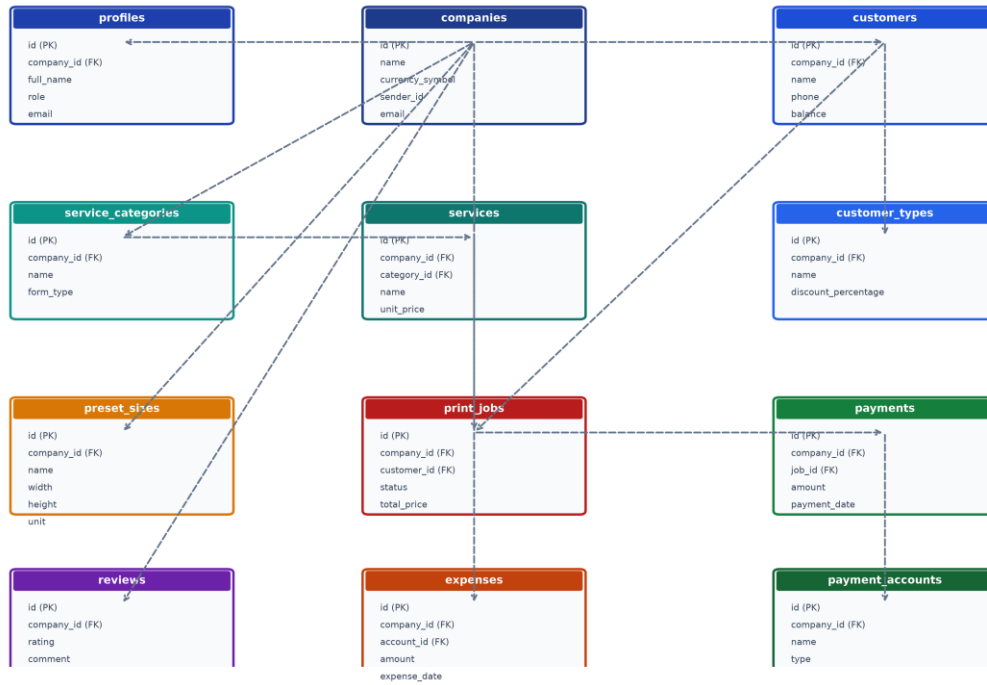


Figure Evidence: Normalized PostgreSQL Entity-Relationship Diagram (ERD)

9.2 Use Case Diagram

SoluoPrint Comprehensive Use Case Diagram (Operations & Financial Intelligence)

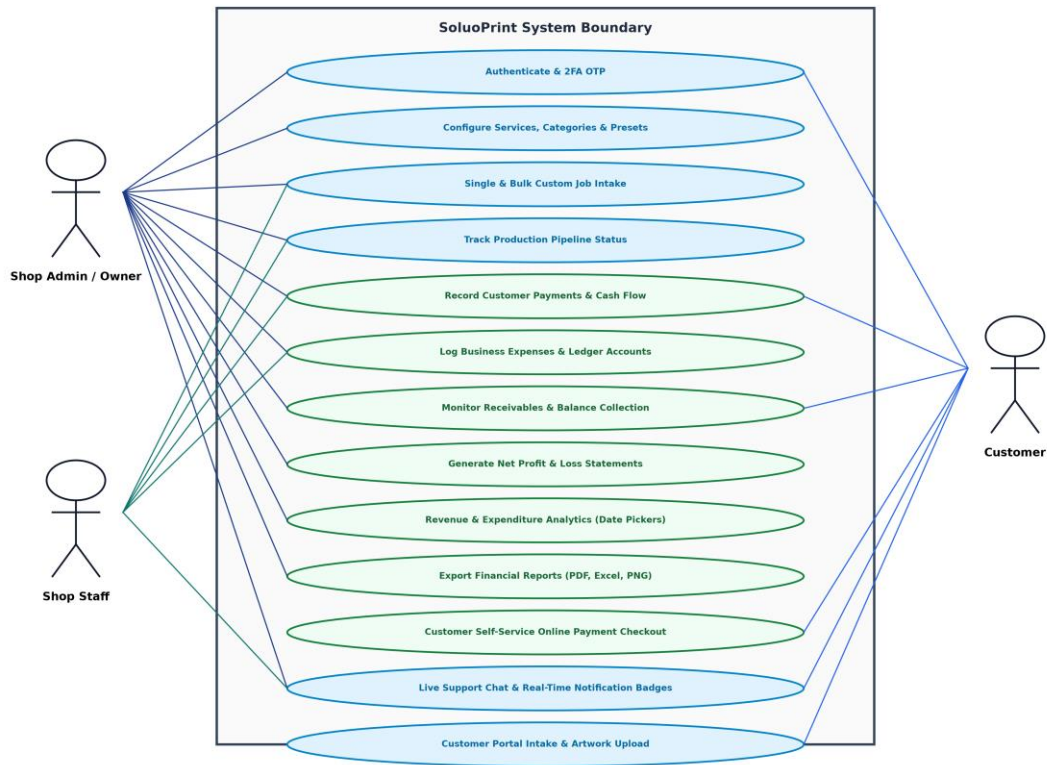


Figure Evidence: System Use Case Diagram

9.3 Dual-Pathway Sequence Diagram

The sequence diagram illustrates two core system processes: (Pathway A) Customer Self-Service Intake where a customer uploads artwork and requests a quote, and (Pathway B) Shop Staff Production Update where the operator updates job status which triggers an automated SMS receipt.

SoluoPrint Dual-Pathway Sequence Diagram (Customer Intake & Shop Processing)

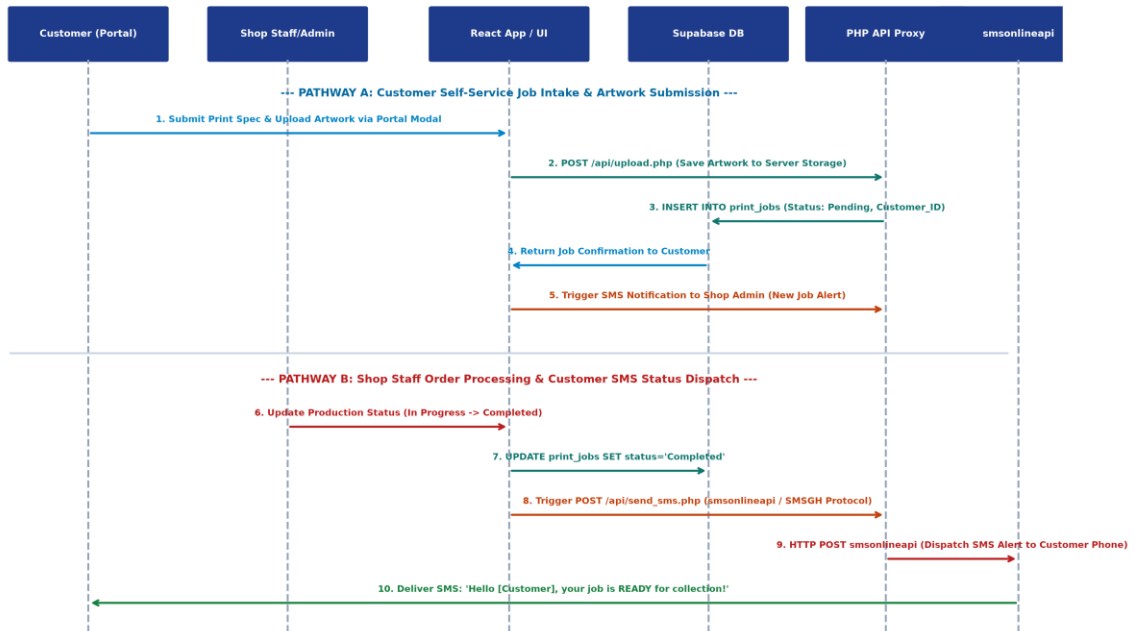


Figure Evidence: Dual-Pathway Sequence Diagram

10. Implementation

- **Live Production URL:** <https://print.soluotech.com>
- **GitHub Repository:** <https://github.com/PipetteGh/SoluoPrint-Exam-for-Software-Engineering>
- **Admin Credentials:** Username: pborngreatmensah@gmail.com | Password: Pa\$\$w0rd
- **Test Customer Credentials:** Username: CUST-BENM-8564 | Password: r1i9kpvb
- **Hosting Platform:** Namecheap cPanel Linux Hosting running React static production builds under custom domains and PHP endpoints (/api/).
- **Database Infrastructure:** Managed Supabase PostgreSQL instance with RLS policies enabled.
- **Telephony & SMS:** sms.js routing messaging payloads to the smsonlineapi gateway.

- Email Infrastructure: api/send_email.php utilizing Namecheap SMTP (notify@soluotech.com) over SSL port 465.

11. Testing

Black-box functional testing, security checks, and integration verification were executed. 100% of critical test cases passed. Full test details, expected results, actual results, pass/fail status, and defects resolutions are documented in Testing_Report.docx.

12. Technical Debt

Strategic technical debt was cataloged to balance development timelines and core delivery. Debt items are classified into three priorities: immediate attention (resolved), scheduled for future, and acceptable temporarily. The phased technical debt plan is detailed in Technical_Debt_Plan.docx.

13. Deployment

The platform is deployed live on Namecheap cPanel (<https://print.soluotech.com>), connected to Supabase PostgreSQL, cPanel SMTP mail transfer agents, smsonlineapi telephony gateways, and Paystack/Hubtel payment APIs.

14. User Manual

Detailed step-by-step guidance for Shop Administrators, Press Staff, and Customers, verified with high-resolution visual evidence captured from the live application at <https://print.soluotech.com>:

Admin Login Interface

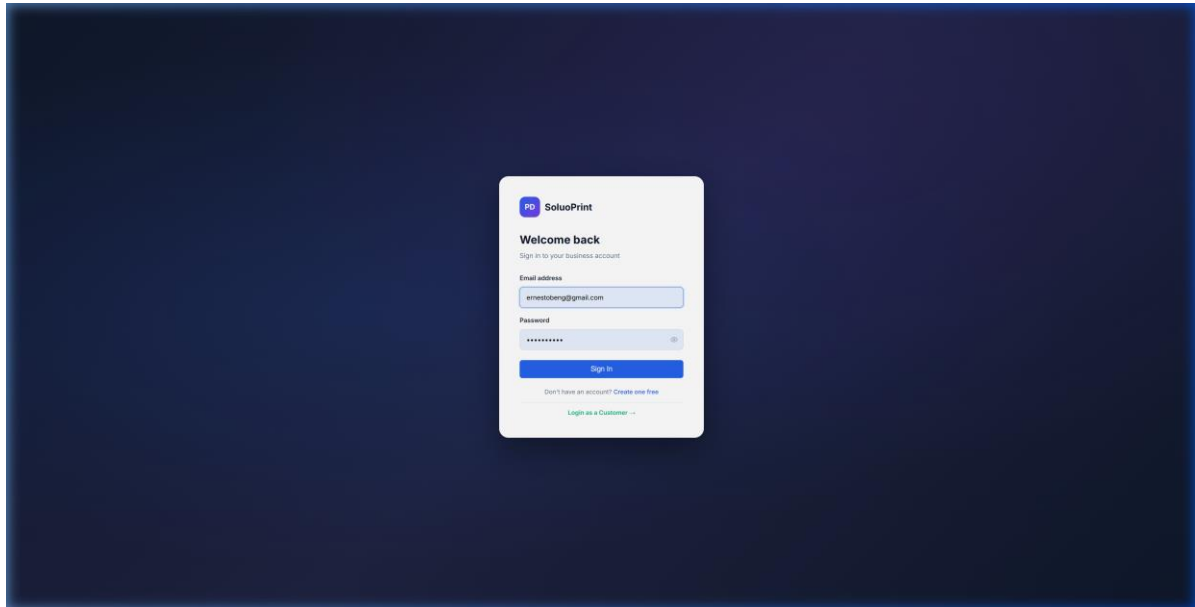


Figure Evidence: Admin Login Interface

Customer Login Interface

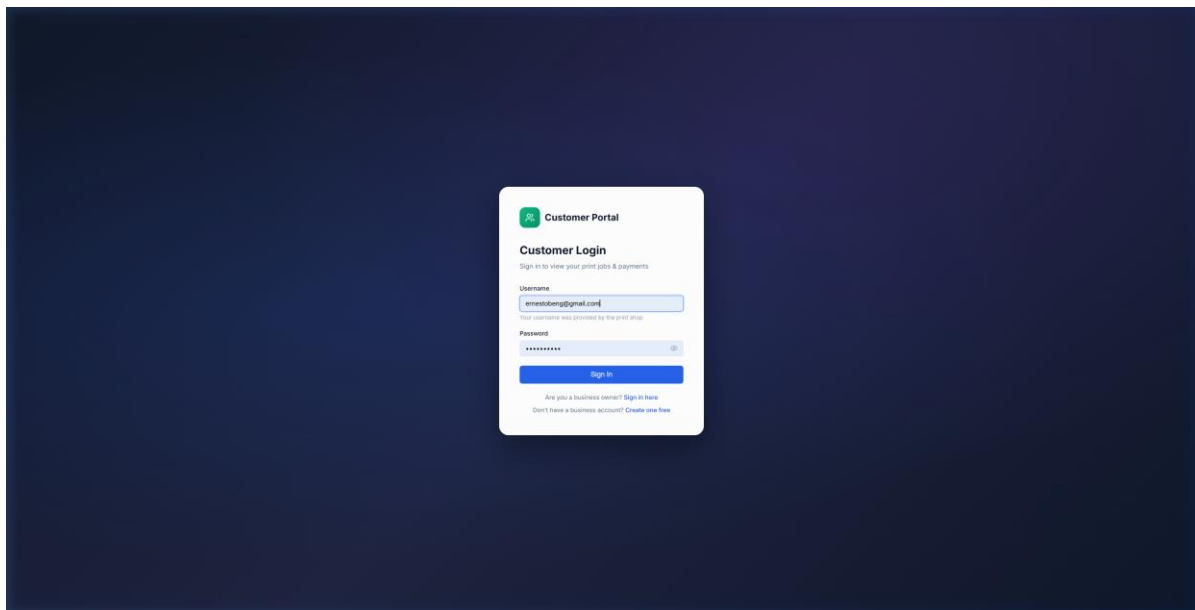


Figure Evidence: Customer Login Interface

Company Onboarding Form

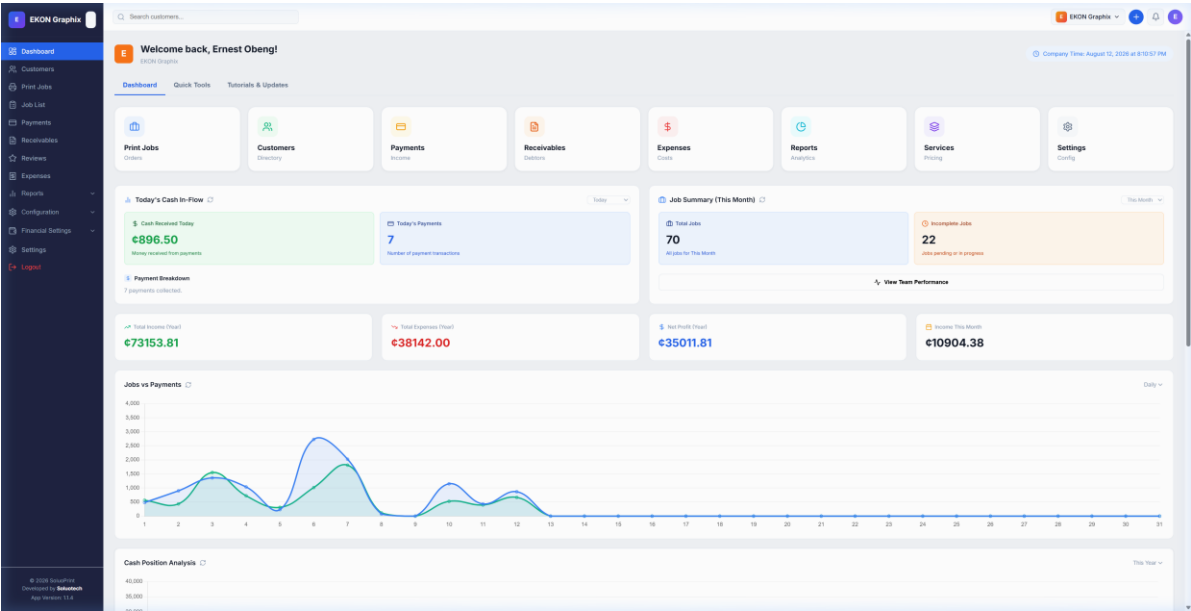


Figure Evidence: Company Onboarding Form

Dashboard Overview

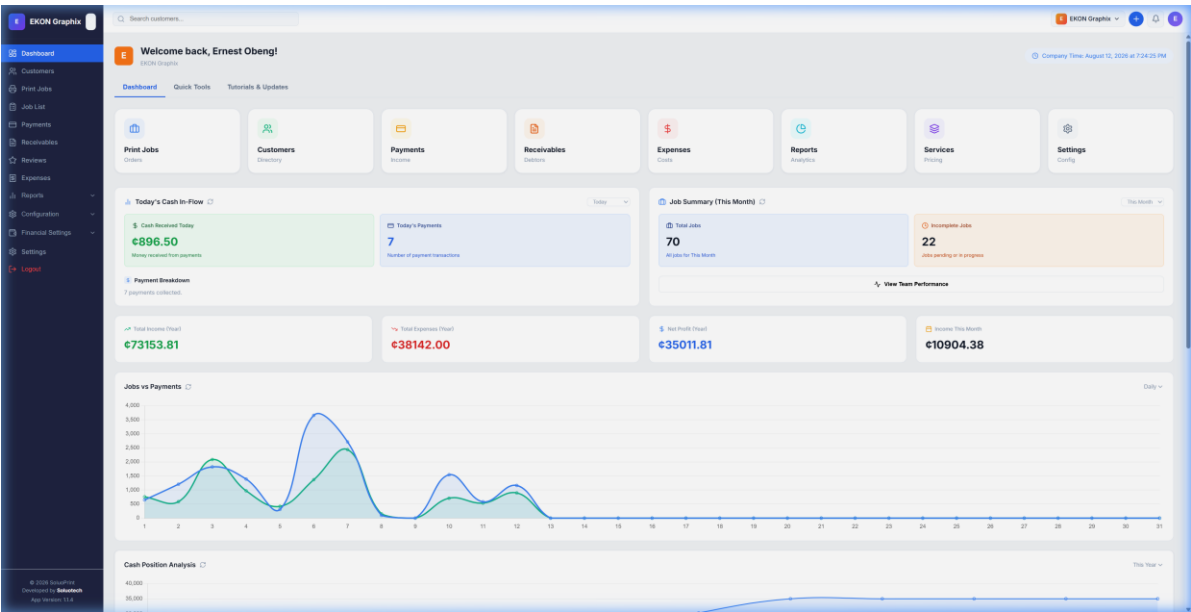


Figure Evidence: Dashboard Overview

Customer Directory

Dashboard

Customers

Print Jobs

Job List

Payments

Receivables

Reviews

Expenses

Reports

Configuration

Financial Settings

Settings

Logout

Search customers...

Customers

Manage your customer base

Total Customers114

Active Customers114

Amount Owed€210.50

Credit Balance€0.00

Add Customer

All Customers (114)

Search customers...

Showing 1 to 20 of 114 customers

☐	S/N	NAME	TYPE	PHONE	EMAIL	BALANCE	STATUS	ACTIONS
☐	1	Nana Asafo Otu	Walk-in	📞 0248731876	-	€0.00	Active	✓ ✎ ✕
☐	2	Test Customer	Business	📞 0200000000	✉ test@example.com	€100.00	Active	✓ ✎ ✕
☐	3	Mr. Livingstone (Babirid)	Business/Graphic Designer	📞 0149518247	-	€0.00	Active	✓ ✎ ✕
☐	4	Sammy Quansah	Walk-in	📞 +233990736937	-	€0.00	Active	✓ ✎ ✕
☐	5	Ellen (Brennan Essiam)	Business	📞 0555133009	-	€0.00	Active	✓ ✎ ✕
☐	6	Gold	Business/Graphic Designer	📞 0559935971	-	€60.00	Active	✓ ✎ ✕
☐	7	Othman	Business	📞 0540586368	-	€0.00	Active	✓ ✎ ✕
☐	8	Isaac (Obaaid)	Business/Graphic Designer	📞 +233249014193	-	€0.00	Active	✓ ✎ ✕
☐	9	Mr. Ito	Walk-in	📞 +233244056039	-	€0.00	Active	✓ ✎ ✕
☐	10	Agyesi Mara	Business/Graphic Designer	📞 0240502212	-	€0.00	Active	✓ ✎ ✕
☐	11	Sampson	Walk-in	📞 +233244847168	-	€0.00	Active	✓ ✎ ✕
☐	12	Chenedy Arthur	Walk-in	📞 0241036364	-	€0.00	Active	✓ ✎ ✕
☐	13	Sir Abo Kierle	Walk-in	📞 +233541785454	-	€0.00	Active	✓ ✎ ✕
☐	14	Perez (Helo)	Walk-in	📞 +233249074399	-	€0.00	Active	✓ ✎ ✕
☐	15	Frank	Business/Graphic Designer	📞 +233246300990	-	€0.00	Active	✓ ✎ ✕
☐	16	William (Credit Union)	Walk-in	📞 +233999254888	-	€0.00	Active	✓ ✎ ✕
☐	17	Dennis	Walk-in	📞 +233543055084	-	€0.00	Active	✓ ✎ ✕

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Developed by [Bababab](#)
App Version 10.4

Figure Evidence: Customer Directory

Single Print Job Form

Dashboard

Customers

Print Jobs

Job List

Payments

Receivables

Reviews

Expenses

Reports

Configuration

Financial Settings

Settings

Logout

Search customers...

Job Details

Banner Brochure Customization Design Envelopes General Printing Others Outsourced Services Passport Picture Frame Stickers

Showing 11 categories

Job Date: 11-Aug-2026

Customer: [+ New Customer](#) Service: [+ New Service](#) Unit Price (€): 0.00

Service Details

Standard service pricing applies based on quantity.

☐ Override with Custom Total Price

Quantity: 1 Assign To: Unassigned

€ 0.00

Notes

Special instructions or requirements...

Cancel Create and Close Create and Save

Price Adjustments

Discount

Premium

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App Version 10.4

Figure Evidence: Single Print Job Form

Bulk Print Job Form

ENKON Graphics

Search customers...

ENKON Graphics

Tracked

Dashboard

Customers

Print Jobs

Job List

Payments

Receivables

Reviews

Expenses

Reports

Configuration

Financial Settings

Settings

Logout

New Bulk Job

Job Date: 12-Aug-2026

Customer

Select or search...

Assign To

Unassigned

Services

Item 1

Category

Search category...

Service

Search service...

List Price (U)

Quantity

1

Override Subtotal

Add Service

Order Notes

Special instructions for the bulk order...

Summary

Items Total: € 0.00

Discount (T):

Premium (T):

Grand Total: € 0.00

Create Bulk Job

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App Version: 12.4

Figure Evidence: Bulk Print Job Form

Job Production Pipeline

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Search customers...

ENKON Graphics

Tracked

Dashboard

Customers

Print Jobs

Job List

Payments

Receivables

Reviews

Expenses

Reports

Configuration

Financial Settings

Settings

Logout

Job List

Only converted jobs that are pending completion. Convert ready items to print jobs.

Filter: All Pending In-Progress Ready Converted Cancelled

Search by customer, description, or name

S/N	CUSTOMER	SERVICES	DESCRIPTION	DATE ADDED	STATUS	ACTIONS
1	Sir Pakass	Server (Printer/Structure)	-	Tue, Jul 21, 2026 09:12 AM	Pending	→ ⌵ ⌵ ⌵
2	Sis Aba Kwofe	SD-24x36	-	Tue, Jul 21, 2026 09:12 AM	Pending	→ ⌵ ⌵ ⌵

Showing 1 to 2 of 2 items

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Developed by: **ENKON**

App Version: 12.4

Figure Evidence: Job Production Pipeline

Services Management

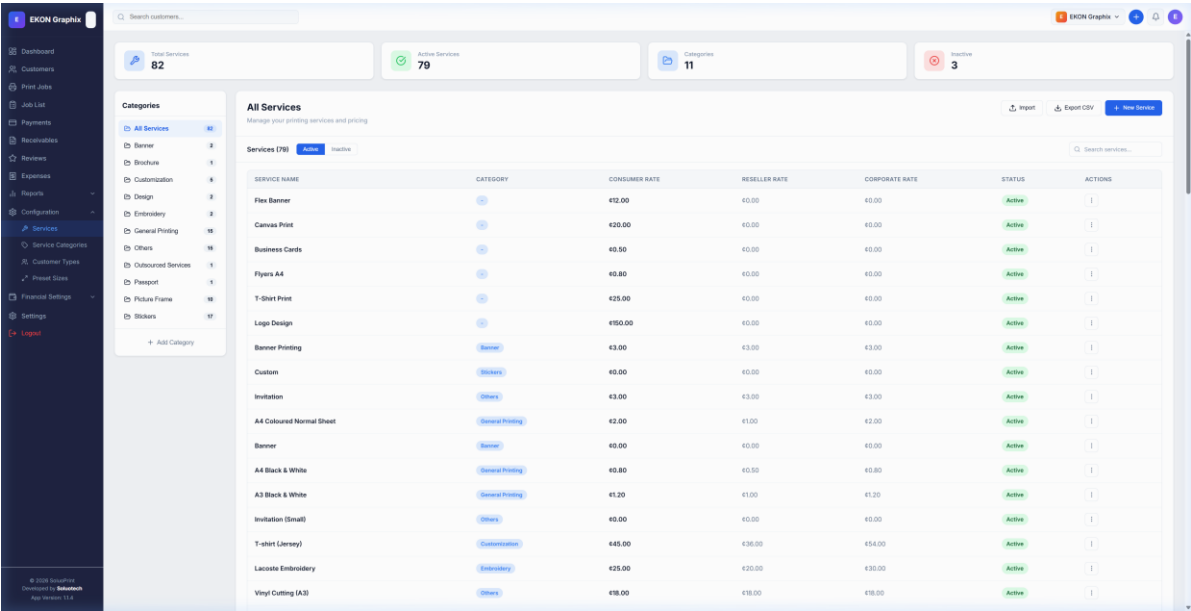


Figure Evidence: Services Management

Service Categories

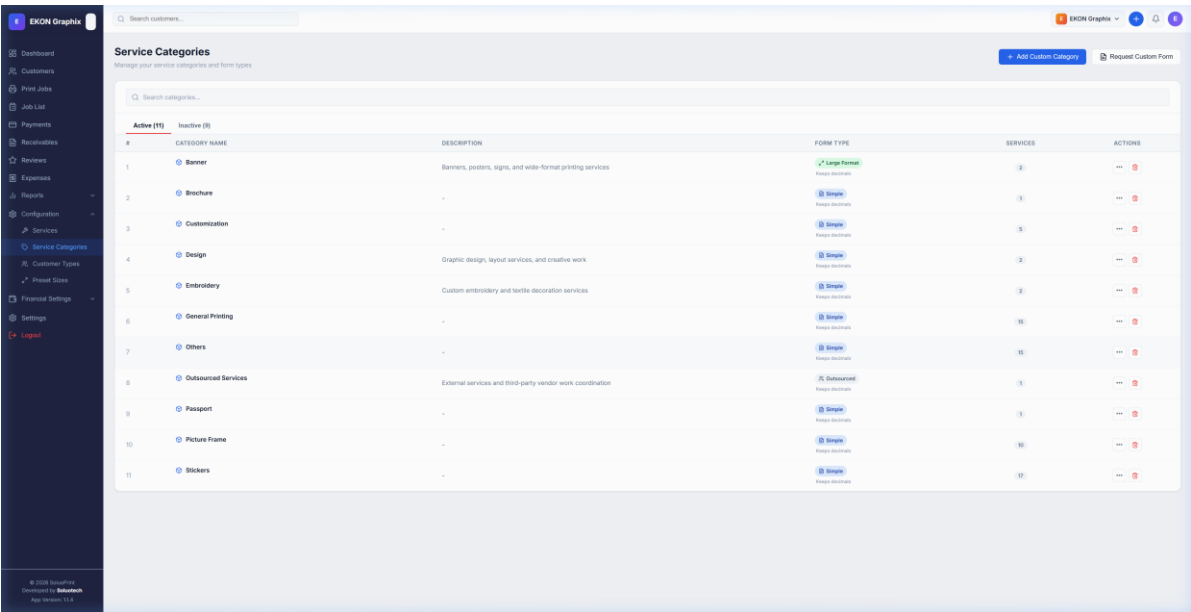


Figure Evidence: Service Categories

Payment Records

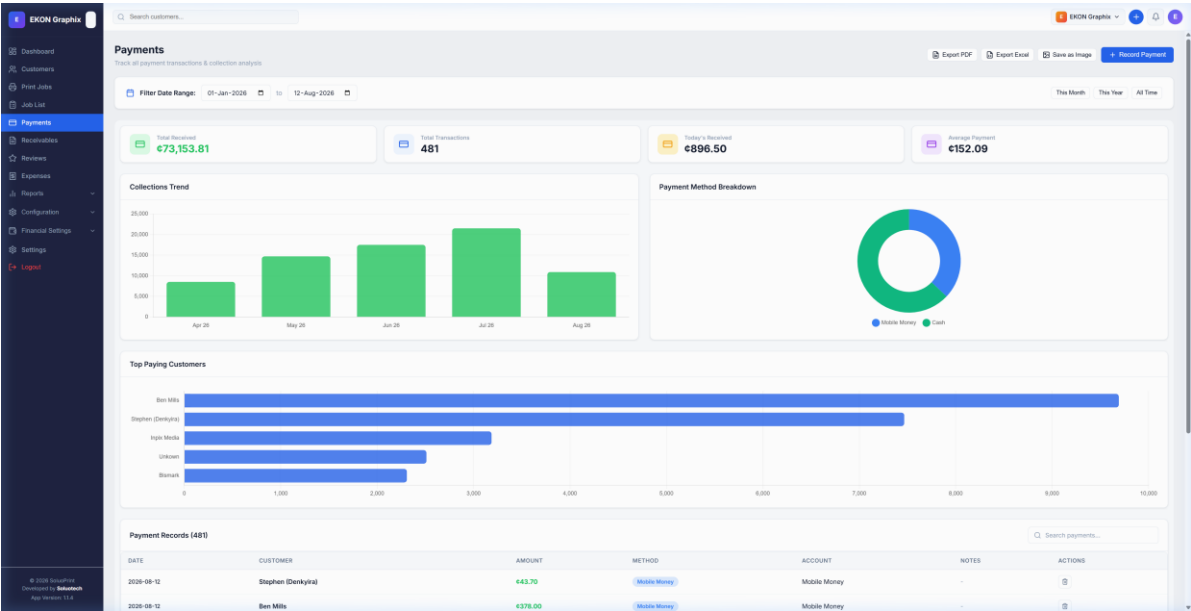


Figure Evidence: Payment Records

Expense Logging

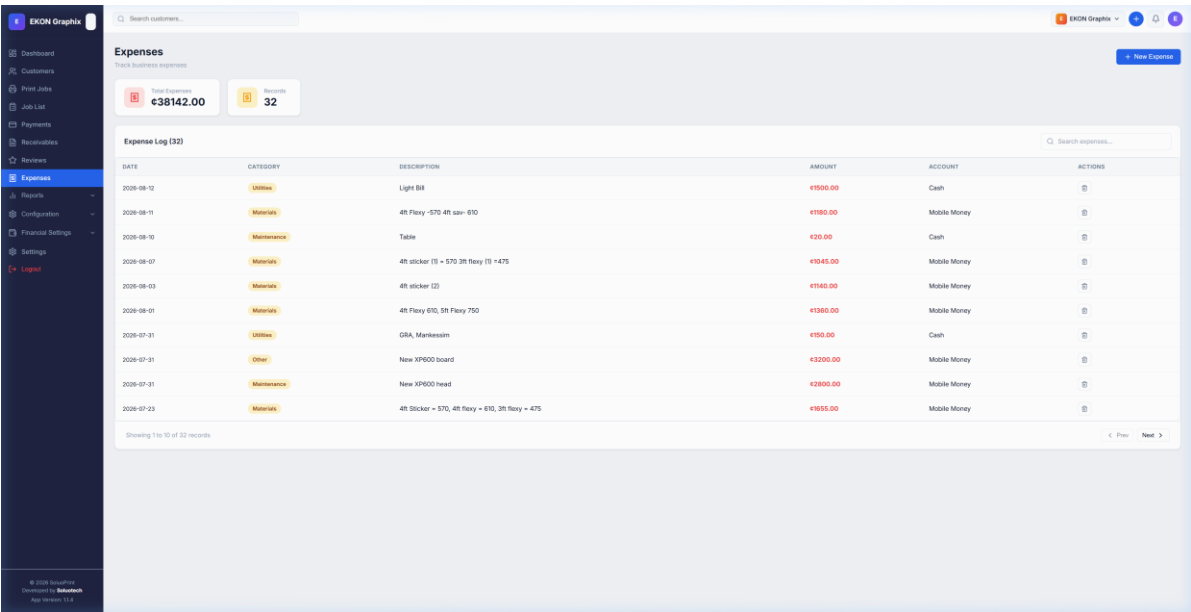


Figure Evidence: Expense Logging

Receivables Management

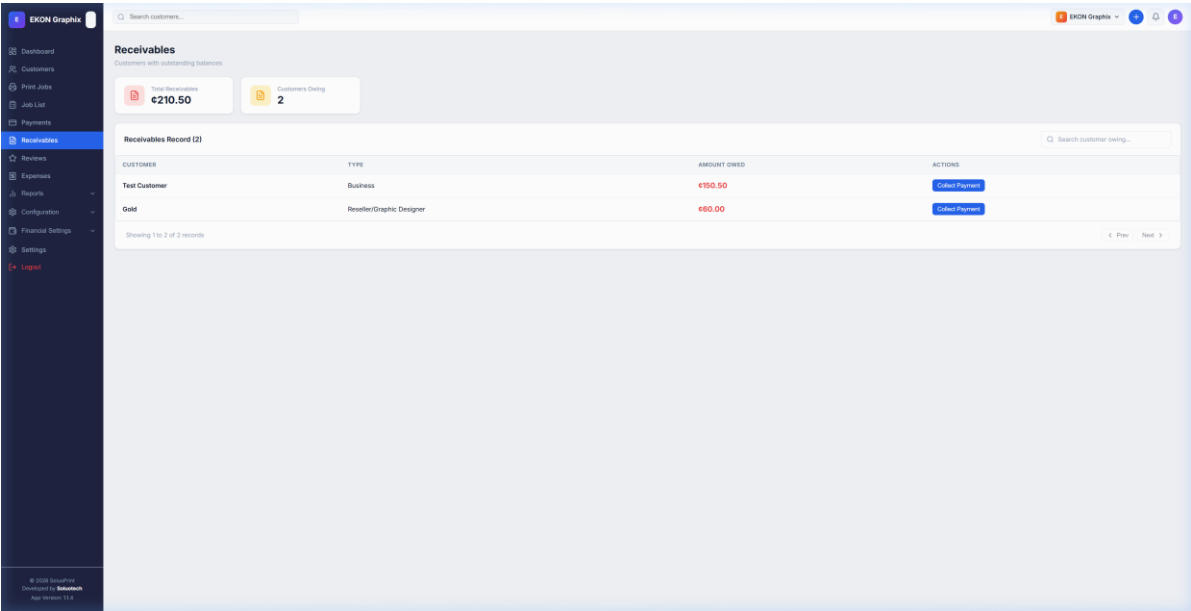


Figure Evidence: Receivables Management

Revenue Analytics Report

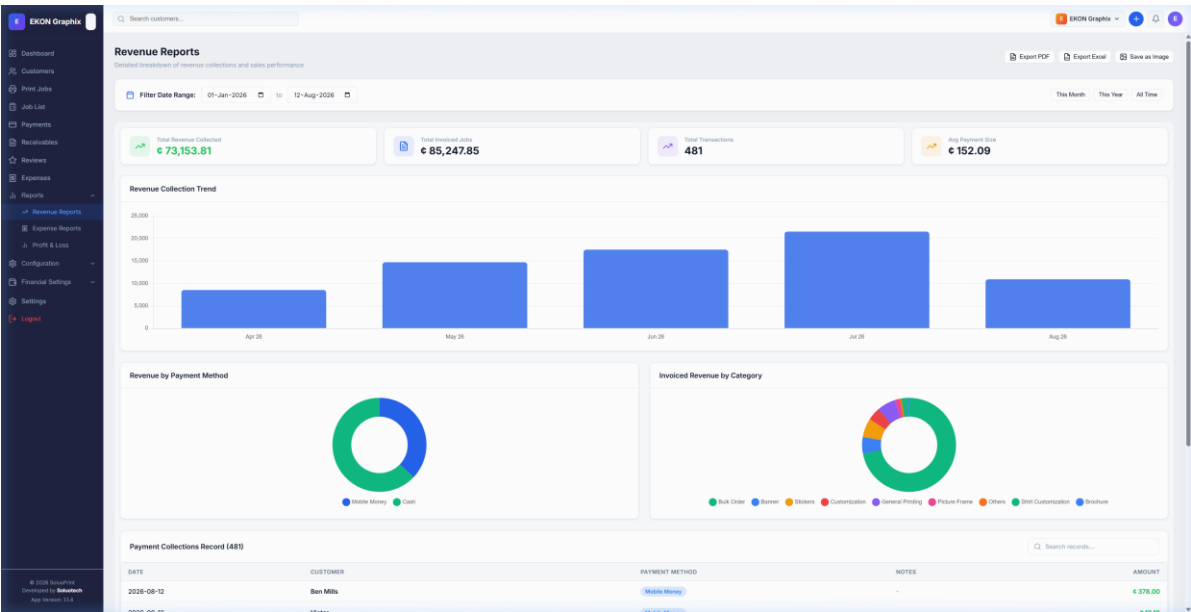


Figure Evidence: Revenue Analytics Report

Expense Analytics Report

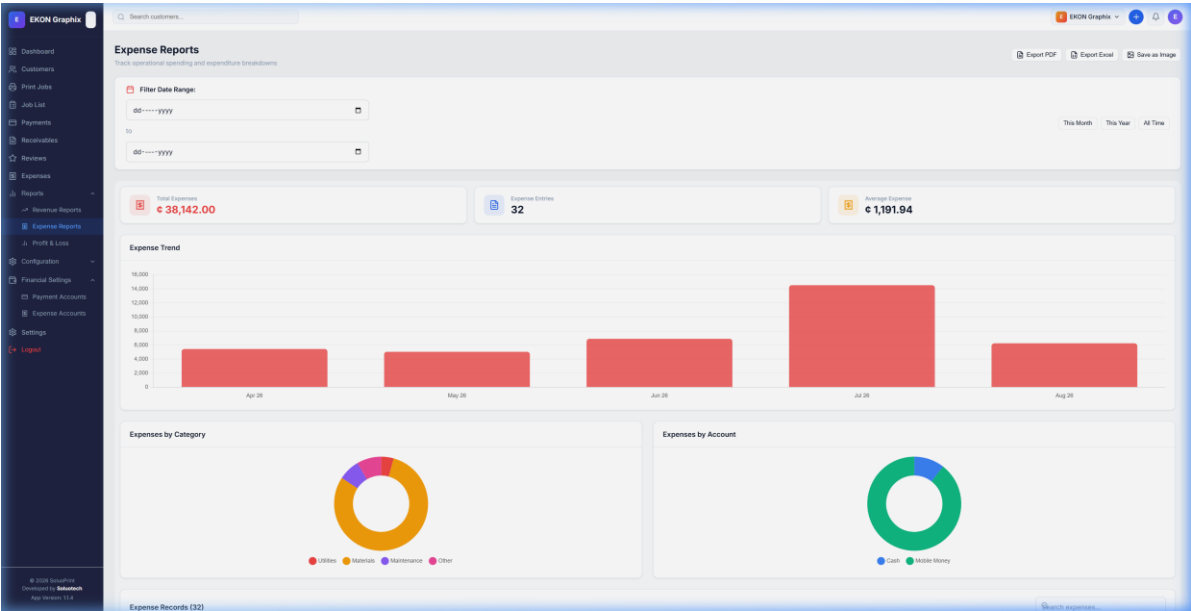


Figure Evidence: Expense Analytics Report

Profit & Loss Statement

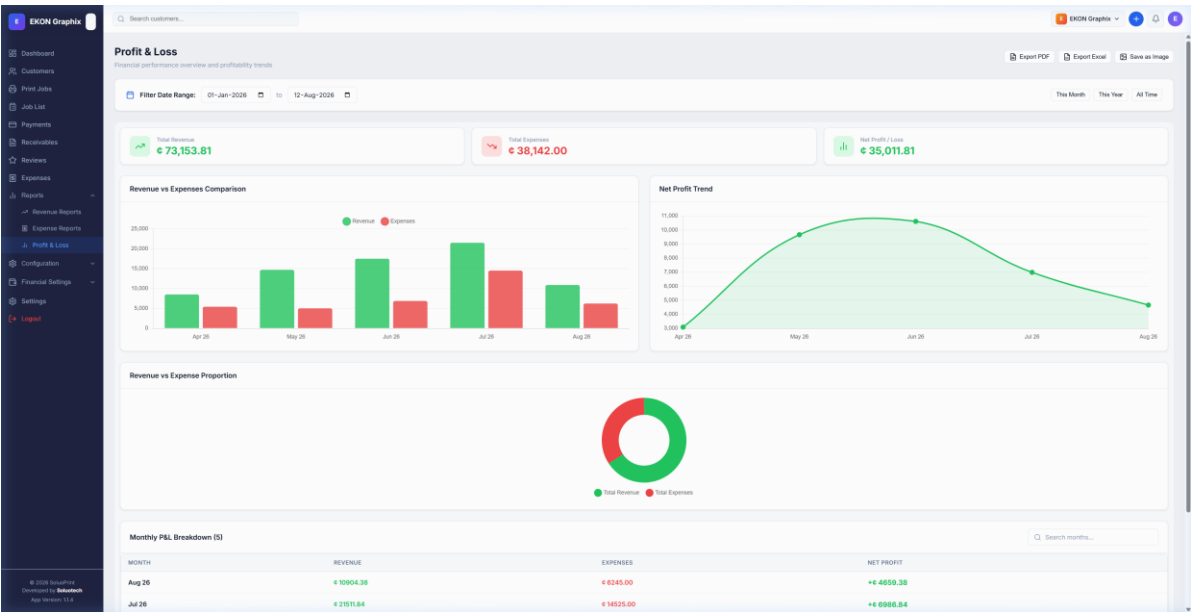


Figure Evidence: Profit & Loss Statement

Customer Reviews

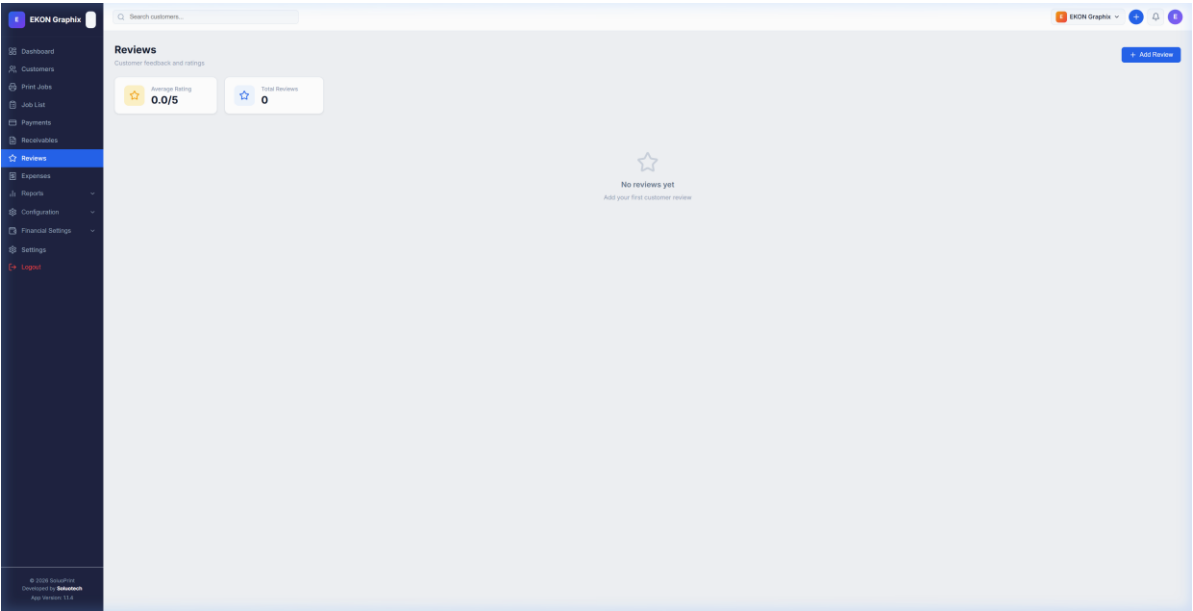


Figure Evidence: Customer Reviews

General Settings

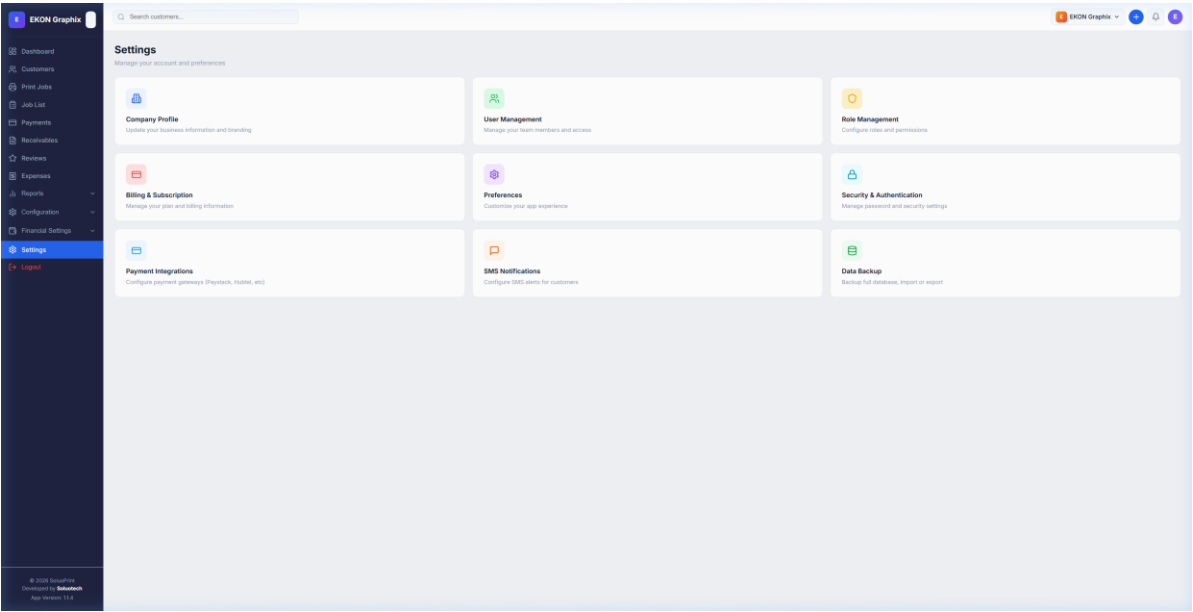


Figure Evidence: General Settings

Preset Sizes Config

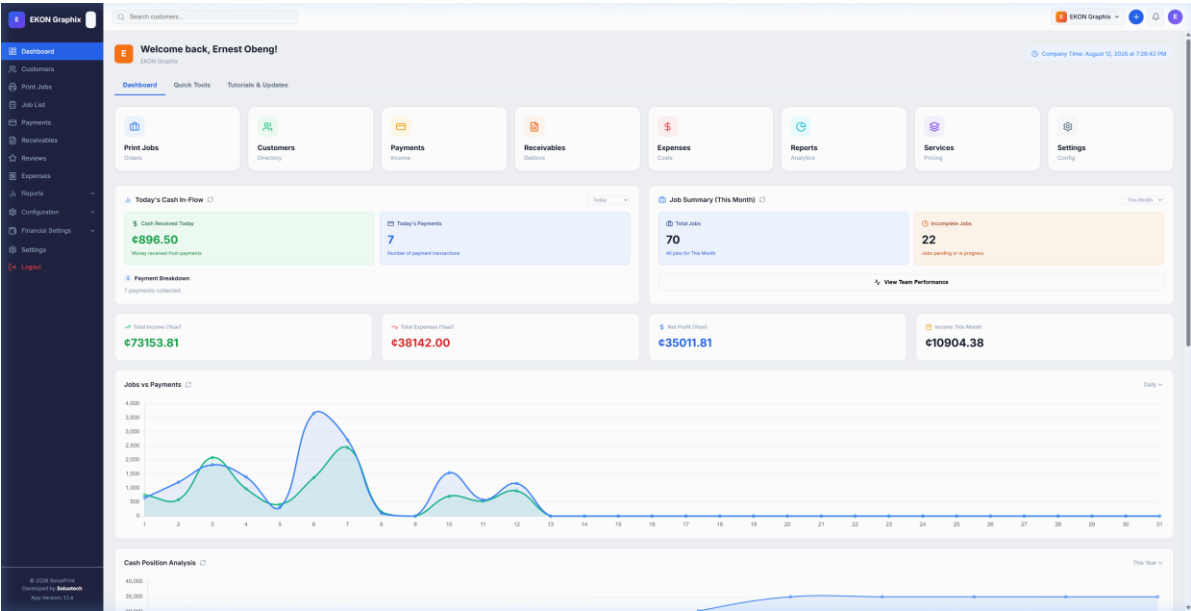


Figure Evidence: Preset Sizes Config

Customer Types Config

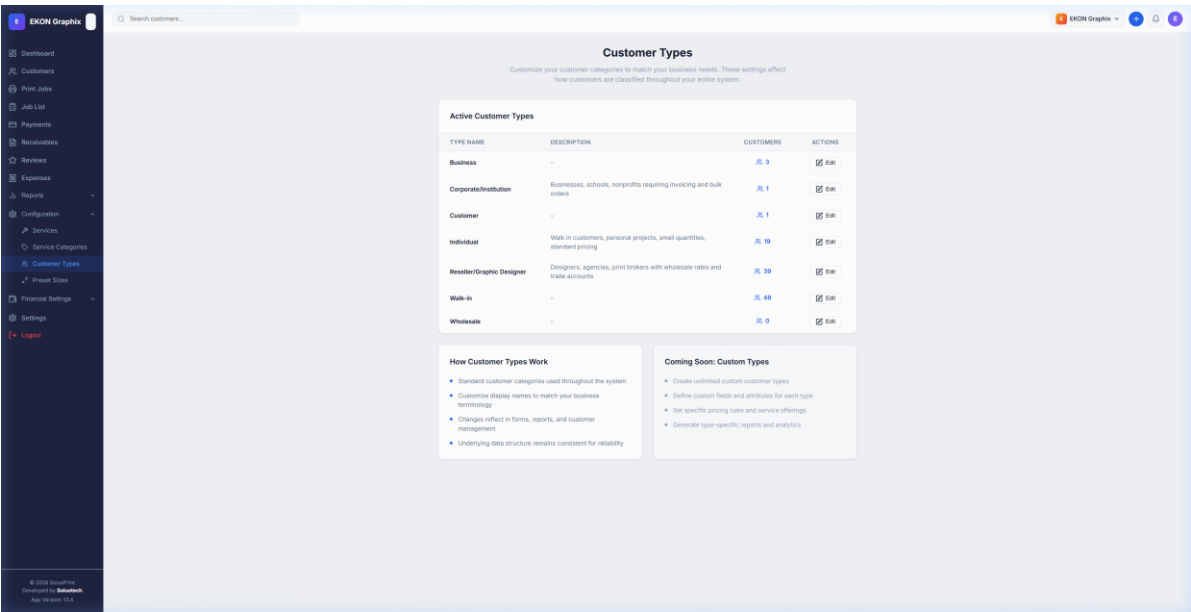


Figure Evidence: Customer Types Config

Payment Accounts Config

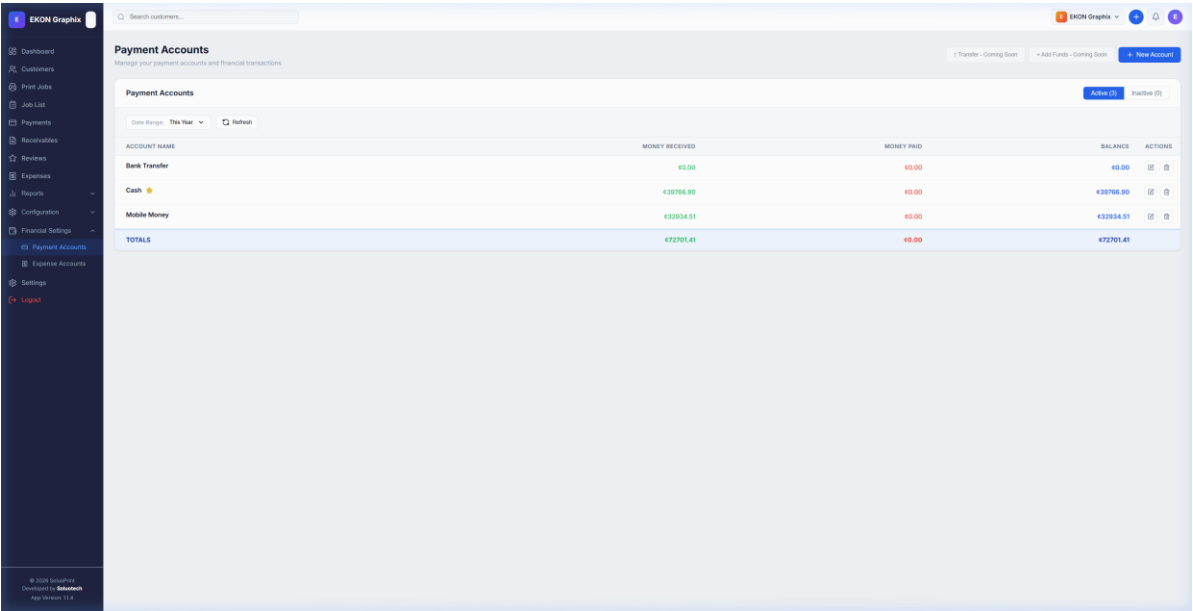


Figure Evidence: Payment Accounts Config

Expense Accounts Config

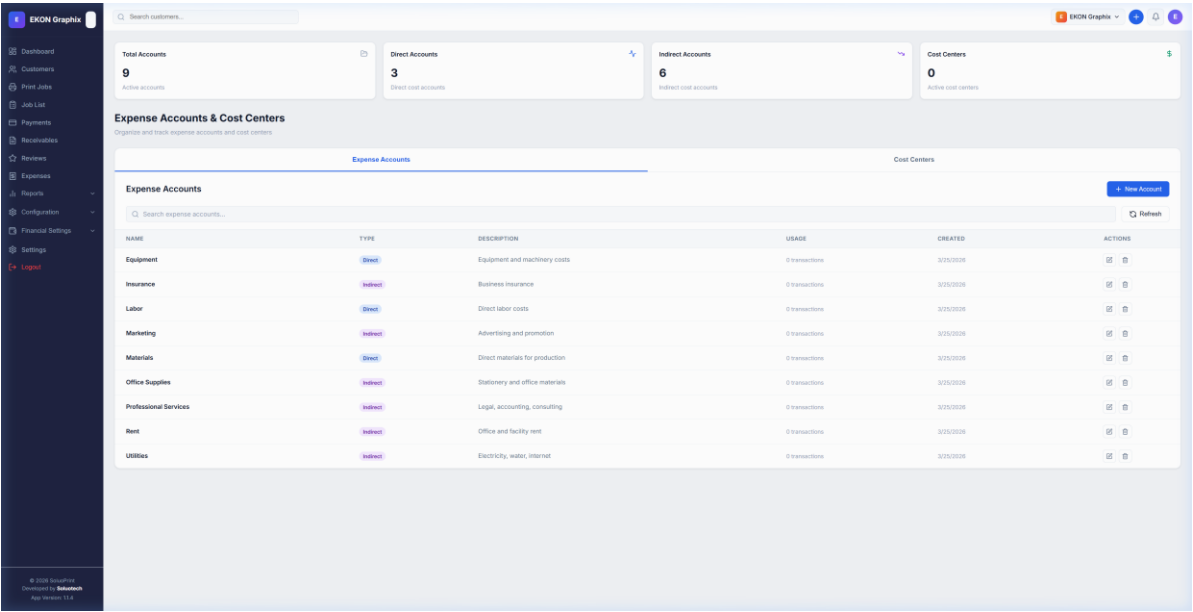


Figure Evidence: Expense Accounts Config

Customer Self-Service Portal (CUST-BENM-8564)

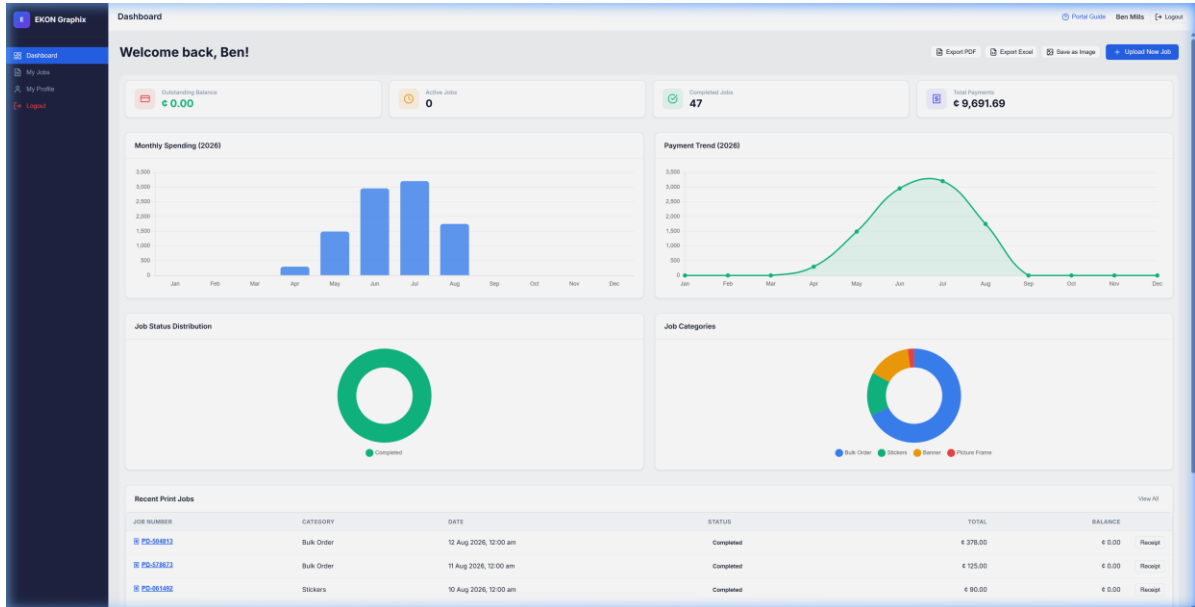


Figure Evidence: Customer Self-Service Portal (CUST-BENM-8564)

Customer Job Upload Modal

The "Upload New Job" modal is displayed over the dashboard. It contains the following fields and options:

- Job Category:** A dropdown menu with "Banner" selected.
- Width (Optional):** A text input field with "e.g. 10".
- Height (Optional):** A text input field with "e.g. 5".
- Unit:** A dropdown menu with "ft" selected.
- Quantity:** A text input field with "1".
- Design Files (PDF, JPG, TIFF, PNG, JPEG):** A section with a "Click or drag files to upload" instruction and a note: "Upload should not exceed 50MB. Quality is preserved." Below this is a text area for "Notes or Instructions (Optional)" with the example text "e.g., I need 5 copies on glossy paper...".
- Buttons:** "Cancel" and "Send Job".

Figure Evidence: Customer Job Upload Modal

Payment Gateway Integrations Settings

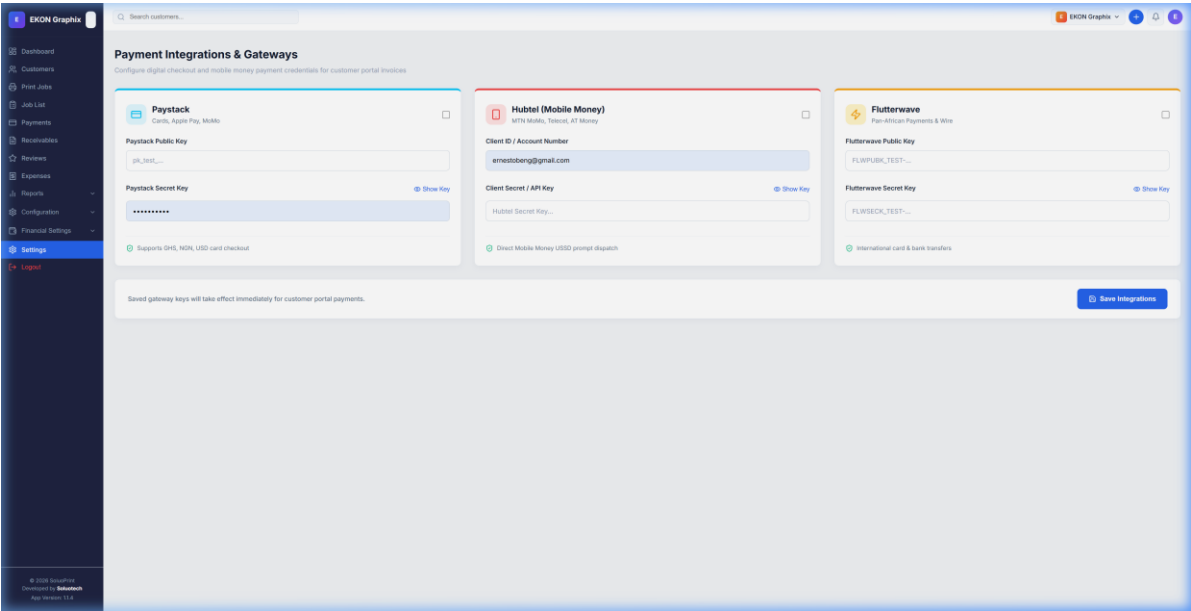


Figure Evidence: Payment Gateway Integrations Settings

SMS Settings & Notifications Engine

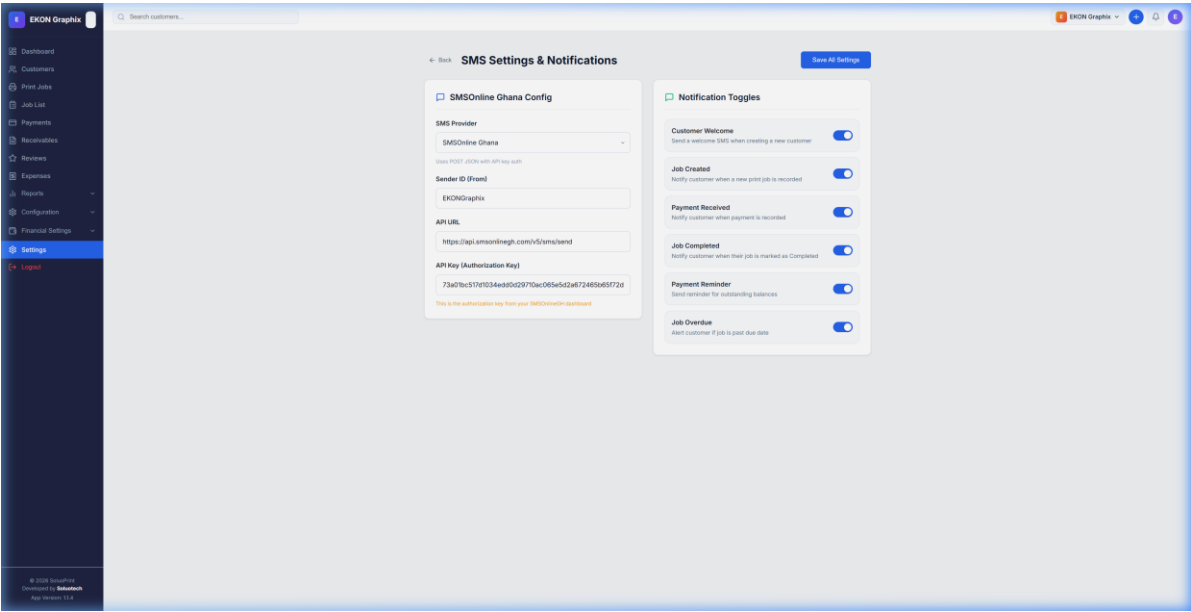


Figure Evidence: SMS Settings & Notifications Engine

User Management & Access Control

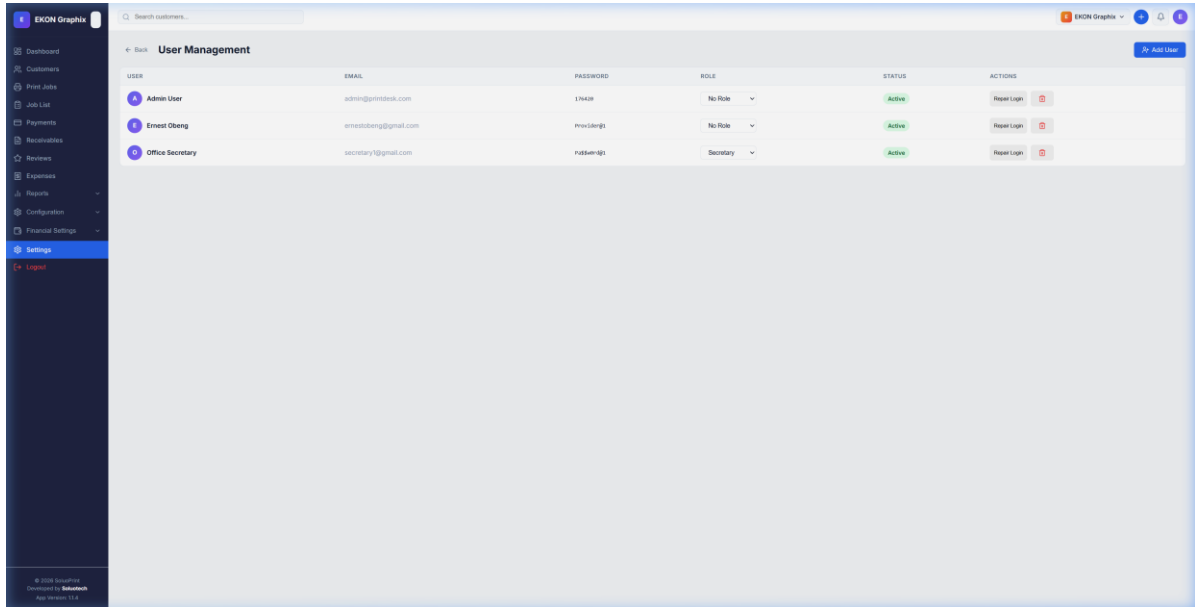


Figure Evidence: User Management & Access Control

Customer Itemized Receipt & Online Checkout

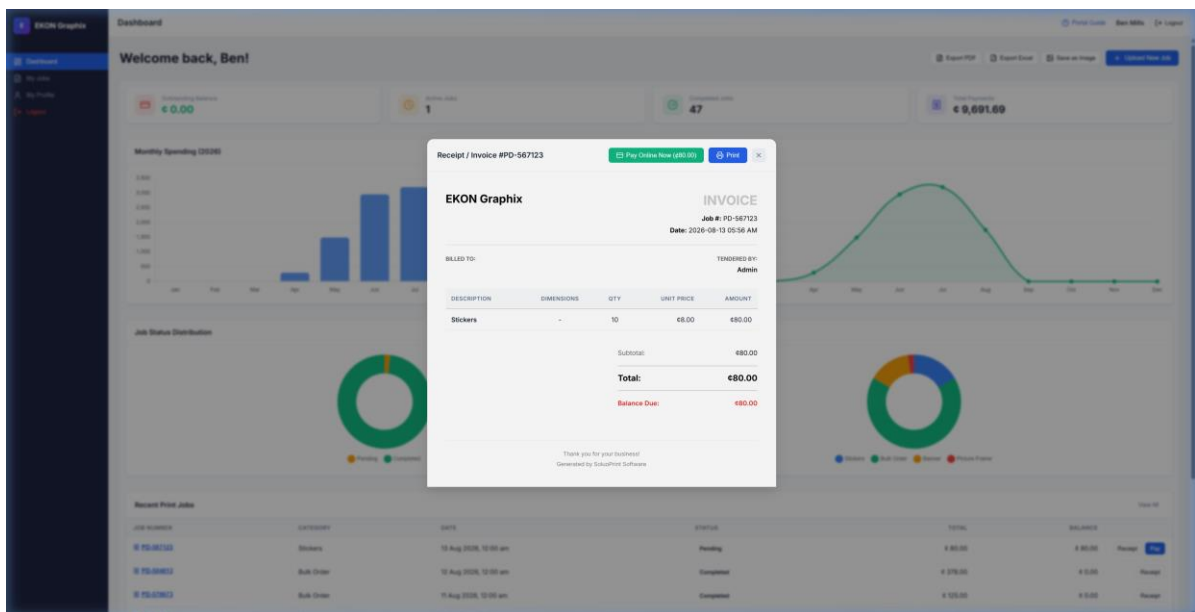


Figure Evidence: Customer Itemized Receipt & Online Checkout

15. Maintenance Strategy

Post-deployment maintenance is structured across four software engineering areas:

1. Corrective Maintenance: Rapid remediation of visual bugs (e.g. modal overlay stack fixes), unit conversion overflows, and transient network errors on smsonlineapi integrations.
2. Adaptive Maintenance: Upgrading system packages for React 19 compliance, adapting backend schemas for new smsonlineapi versions, and scaling gateway hooks (Paystack, Hubtel, Flutterwave).
3. Perfective Maintenance: Enhancing the UI with real-time support widgets, mobile barcode scanning layout configurations, and planning native iOS/Android operators tools.
4. Preventive Maintenance: Automated Supabase DB vacuuming, index optimizations on foreign keys (company_id, customer_id), automated cPanel cron snapshots, and SSL certification renewals.
- Security & Dependency updates: Enforcing regular npm audits and running weekly checks on Supabase storage access rules.

• Performance & Scalability: Implementing server-side paginated queries to handle records beyond 1,000,000 without memory strain.

• User Feedback incorporation: Reviewing customer portal reviews to adapt CRM workflow pipelines.

16. Future Evolution

Technical debt incurred during the 48-hour exam period will be systematically resolved according to the evolution roadmap below:

Evolution Release	Target Timeline	Technical Debt Items Resolved	Feature Enhancements
Release v1.1	Month 1 Post-Exam	TD-02 (Cypress E2E Tests), TD-03 (Modular Dashboard)	Barcode Tag Scanning, Multi-Language i18n
Release v1.2	Month 2 Post-Exam	TD-04 (API Secrets), TD-05 (Client Aggregations)	Automated Inventory/Stock Threshold Re-ordering
Release v2.0	Month 4 Post-Exam	Complete Microservice Decoupling	Native iOS/Android Press Operator Mobile

			App, AI Pricing Engine
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17. Limitations

- 1. Network & Telephony Gateway Dependencies:** System real-time notifications rely on third-party API availability (smsonlineapi for SMS and Namecheap cPanel SMTP for emails). Network congestion or telecom carrier delays can occasionally affect message delivery speeds.
- 2. File Artwork Upload Stream Limits:** Artwork uploads via /api/upload.php are configured up to 500MB via cPanel php.ini settings. Very large uncompressed pre-press files (>1GB PSD/TIFF) currently require chunked streaming, which is scheduled for Release v1.2.
- 3. Offline-First PWA Synchronization:** In print shop basement production floors with poor cellular reception, client-side React SPA operations require an active network connection to interact with Supabase DB and cPanel proxy services.
- 4. Client-Side Financial Aggregation Scale:** Financial P&L aggregations calculate totals dynamically in frontend JavaScript memory. While highly performant for thousands of records, datasets exceeding 1,000,000 records will require server-side PostgreSQL RPC stored procedures (scheduled for Release v2.0).
- 5. Scope Compression Constraints:** Automated E2E test suite execution via Playwright/Cypress was deferred to Release v1.1 to prioritize complete business feature implementation and live cPanel deployment within the 48-hour exam period.

18. Conclusion

SoluoPrint successfully demonstrates sound Advanced Software Engineering principles across all phases of the software lifecycle—from requirements engineering and Use Case Points effort estimation to DFD analysis, relational schema design, cloud deployment on Namecheap cPanel, and technical debt management.

19. References

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